Cultural-QLoRA: Hofstede-Conditioned Persona Adaptation for Small Korean LLMs

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Abstract

Off-the-shelf instruction-tuned LLMs over-represent Anglo cultural priors even when prompted in Korean, limiting their utility for non-Anglo applications such as consumer-persona generation and culturally grounded content. We investigate whether parameter-efficient cultural conditioning—QLoRA fine-tuning on culturally-grounded data (CultureBank, Nemotron-Personas-Korea) with Hofstede-6D system-prompt conditioning—can measurably shift a small (3B/7B) instruction-tuned model's response distribution toward a target culture's empirical opinion distribution. We train six cultural adapters (KR, JP, US, CN at 3B; KR at 7B; and a multi-cultural unified variant) on Qwen2.5-Instruct backbones and evaluate against (i) four Korean benchmarks (KoBBQ, KMMLU, HAE-RAE, CLIcK), (ii) cross-cultural alignment via Kolmogorov–Smirnov distance against GlobalOpinionQA empirical distributions plus BLEnD MCQ accuracy, and (iii) a 3-judge LLM panel (GPT-5.5, Claude Opus 4.7, MiMo v2.5-Pro) scoring cultural authenticity. Our key finding is that QLoRA-based cultural reweighting on a 3B model produces measurable distributional shifts toward the target country on multinational survey questions, and transfers across cultures via a single multi-cultural adapter conditioned on a culture token. We also surface an honest asymmetry: while Cultural-JP/US/CN adapters reduce KoBBQ bias by 18–24pp, the Cultural-KR adapter shows the smallest debiasing effect (−4.5pp: 0.450 to 0.405), suggesting that training on target-culture data may amplify in-culture stereotype patterns. We frame this as an alignment–debiasing asymmetry distinct from generic bias mitigation. Code, adapters, decision log, and reproducibility scripts are released.

1 Introduction

Large language models (LLMs) increasingly mediate decisions in consumer modeling, persona generation, and cross-cultural product design. A growing body of evidence shows these models embed predominantly Anglo-American cultural priors Durmus et al. [2023], Cao et al. [2023], which presents a structural problem for non-Anglophone users. Even when prompted in Korean, Vanilla Qwen2.5-7B-Instruct scores only 32% on KMMLU Korean-History and 47% on the HAE-RAE Bench 1.1, while exhibiting a 35% KoBBQ bias rate driven by Anglo-default framings of Korean social contexts.

Prior work either (a) measures the cultural-bias gap without remediation Durmus et al. [2023], Myung et al. [2024], Shi et al. [2024] or (b) aligns models via full RLHF on cultural preferences Li et al. [2024], AlKhamissi et al. [2024], which is impractical for small-team deployments. We investigate a middle path: **parameter-efficient cultural conditioning** via QLoRA Dettmers et al. [2023] with Hofstede-6D Hofstede et al. [2010] system-prompt conditioning on culturally-curated training data.

Our contributions are:

1. **Cultural-QLoRA pipeline + 6 trained adapters**: 4-bit NF4 + LoRA (rank 16–32) on Qwen2.5-3B/7B with Hofstede-6D system prompts; per-culture KR/JP/US/CN, plus KR-7B and a multi-cultural unified variant (Run-M) using a `<<culture:xx>>` token (§3).
2. **Cross-cultural alignment evaluation:** GlobalOpinionQA KS distance (150 questions $\times$ 6 samples) and BLEnD MCQ, plus Hofstede dimension ablation (IDV-only/UAI-only/full-6D) and a 3-judge LLM panel (GPT-5.5, Claude Opus 4.7, MiMo) (§4).
3. **Honest asymmetry finding:** Cultural-KR adapter exhibits the smallest debiasing among cultural adapters (−4.5pp KoBBQ bias drop vs. −18 to −24pp for JP/US/CN), suggesting that target-culture training data can amplify in-culture stereotype patterns—a tradeoff distinct from generic bias mitigation.
4. **Open-source release**: code, 6 trained adapters, decision log (17 documented judgments), audit-traced bug fixes, and reproducibility scripts.

Section 2 reviews prior work; §3 details our approach; §4 reports quantitative results; §5 provides qualitative analysis; §6 concludes with limitations.

2 Related Work

Cultural alignment in LLMs. Durmus et al. [2023] introduced the GlobalOpinionQA benchmark, showing that current LLMs systematically over-align with Western populations on Pew/World Values Survey items. Cao et al. [2023] extended this with multilingual probes. AlKhamissi et al. [2024] found that explicit cultural prompting partially shifts model outputs but does not modify model weights. Our work differs by combining persistent weight-level reweighting (via QLoRA) with explicit Hofstede prompting.

Korean NLP benchmarks. We evaluate against four Korean benchmarks: KoBBQ Jin et al. [2024] (Korean Bias Benchmark for QA, 76K samples with Korean-specific stereotype categories), KMMLU Son et al. [2024] (Korean MMLU spanning 45 subjects), HAE-RAE Bench 1.1 Son et al. [2023] (Korean cultural and linguistic knowledge), and CLIcK Kim et al. [2024] (Korean Cultural and Linguistic Intelligence). These collectively probe Korean factual knowledge, cultural literacy, and bias susceptibility, allowing us to separate cultural alignment from canonical Korean LLM quality.

Cross-cultural and persona generation datasets. Shi et al. [2024] introduced CultureBank, a community-sourced repository of cultural descriptors with country tags. Myung et al. [2024] released BLEnD, a country-conditioned cultural common-sense MCQ benchmark. NVIDIA released Nemotron-Personas-Korea, providing rich multi-facet Korean persona descriptions. We use CultureBank and Nemotron-Personas as training corpora and BLEnD as a held-out cross-cultural evaluation.

Parameter-efficient adaptation. QLoRA Dettmers et al. [2023] combines 4-bit NF4 quantization with Low-Rank Adapters Hu et al. [2022]. We use rank 16 (attention-only) and rank 32 (all-linear) configurations.

Comparison to MiroFish and prior cultural-simulation work. The MiroFish project Ji et al. [2026] (62K-star repository) and OASIS Yang et al. [2024] approach cultural modeling via multi-agent *simulation*: instantiating LLM personas with culture-tagged scaffolds and observing emergent behavior. Our approach is distinct in three ways. First, we

Table 1: Per-culture training set sizes (instruction-format rows).

Culture	CultureBank	Nemotron-Personas	Total
KR	1,026	4,800	5,826
JP	916	0	916
US	3,600	0	3,600
CN	740	0	740

modify model weights via QLoRA so cultural alignment is a property of the deployed artifact, not a prompt-time scaffold; this enables sub-second inference vs. simulation overhead. Second, we evaluate against *empirical distributions* (GlobalOpinionQA per-country survey responses) rather than agent self-report. Third, our 4-culture transfer matrix (Table 4) directly tests whether cultural reweighting is destructive of other-culture capability—a question MiroFish-style simulation cannot answer because its agents are unmodified. Concurrent CultureLLM Li et al. [2024] also fine-tunes for cultural conditioning but on synthetically-augmented data; we use community-curated CultureBank as our cultural source, yielding 6,282 KR rows from human-flagged TikTok and Reddit cultural descriptors.

3 Approach

3.1 Problem formulation

Given a target culture c with Hofstede-6D vector $h_c \in \mathbb{R}^6$, a base LLM M_θ, and an input prompt p in language ℓ_c, we seek to learn an adapter $\Delta\theta$ such that the conditional response distribution $P_{M_{\theta+\Delta\theta}}(\cdot \mid p)$ aligns with the empirical target-culture distribution $P_c^\star(\cdot \mid p)$ measured via multinational survey responses. Our primary alignment metric is the discrete Kolmogorov–Smirnov distance:

$$D_{\mathrm{KS}}(P, Q) = \max_{i \in \{1,\ldots,K\}} \left| \sum_{j \le i} P_j - \sum_{j \le i} Q_j \right|, \tag{1}$$

computed on the discrete option distribution (option count K varies, $2 \le K \le 9$). Lower D_{KS} indicates closer alignment to the empirical target-country distribution.

3.2 Hofstede-6D system-prompt conditioning

Each training example is prefixed with a system message encoding Hofstede's six dimensions (PDI, IDV, MAS, UAI, LTO, IVR). For Korea: PDI=60, IDV=18, MAS=39, UAI=85, LTO=100, IVR=29. Format (uniform across cultures):

```
You are an AI persona reflecting {country} cultural context.
Hofstede 6D: PDI={p}, IDV={i}, MAS={m}, UAI={u}, LTO={l}, IVR={v}.
Respond authentically from this cultural perspective in {lang}.
```

3.3 Cultural training data

We construct per-culture instruction-tuning corpora from two sources: (i) **CultureBank** Shi et al. [2024] filtered by country tag, with each row producing two instruction variants (descriptor explanation; persona+question pair), and (ii) **Nemotron-Personas-Korea** (KR only), producing three variants per persona (summary, cultural background, lifestyle facet). Per-culture sizes are reported in Table 1.

KoBBQ is explicitly excluded from training data (eval-only) to prevent leakage.

3.4 QLoRA training

4-bit NF4 + bf16 compute; LoRA rank $r \in \{16, 32\}$, $\alpha = 2r$, dropout 0.05; attn-only or all-linear modules. Cosine LR 2e-4, warmup 3%, batch 1 with grad-accum 8, paged AdamW 8-bit, gradient checkpointing. 1–5 epochs depending on dataset size.

3.5 Cross-cultural alignment evaluation

For each (adapter a, target culture c) pair, we evaluate on (i) GlobalOpinionQA Durmus et al. [2023]: 150 questions, 6 model samples per question, computing D_{KS} between the empirical model distribution and the country's empirical distribution from the dataset's `selections` field; (ii) BLEnD MCQ Myung et al. [2024]: country-conditioned cultural common-sense, 80 questions, country-tagged choices.

3.6 Hofstede dimension ablation

We train three KR variants varying only the dimensions encoded in the system prompt: (a) IDV-only (collectivism dim, KR=18), (b) UAI-only (KR=85), (c) full-6D (default). All other hyperparameters and training data are held fixed. Comparing D_{KS} across variants identifies which dimension drives the alignment shift.

3.7 Multi-cultural unified adapter

A single adapter (Run-M) is trained on a mix of all four cultures (10,511 rows = 5,826 KR + 916 JP + 3,600 US + 740 CN) with a culture token `<<culture:xx>>` prepended to each user instruction. Same hyperparameters as per-culture adapters (rank 32 all-linear, 1 epoch, paged AdamW 8-bit). At inference, the requested culture is set via this token; evaluation reports KS for each of the four culture tokens (Table 4).

3.8 CAS LLM-judge panel

A 3-judge panel rates persona authenticity, consistency, and factual accuracy (1–5 each) for 60 generated persona prompts per adapter. Judges are GPT-5.5 (reasoning_effort=none), Claude Opus 4.7, and MiMo v2.5-Pro via the Xiaomi Plan endpoint. Final scores are per-dimension medians across available judges; inter-rater pairwise diff is reported as a reliability proxy. Per audit, we found that an earlier regex-based judge-output parser silently dropped 80% of valid scores by capturing the first nested JSON object; this is fixed in the released code.

4 Experiments

4.1 Data

Korean evaluation benchmarks: KoBBQ [Jin et al., 2024] (200 samples after audit-fix re-run, balanced ambiguous/disambiguated context), KMMLU [Son et al., 2024] (100 samples), HAE-RAE Bench 1.1 [Son et al., 2023] (4 subjects × 25 samples), CLIcK [Kim et al., 2024] (100 samples, Korean cultural literacy). Cross-cultural: GlobalOpinionQA [Durmus et al., 2023] (150 questions per culture, 6 model samples per question), BLEnD MCQ [Myung et al., 2024] (80 per culture, country-conditioned prompting). Training corpora: CultureBank [Shi et al., 2024] country-filtered + Nemotron-Personas-Korea (KR only) as detailed in Table 1.

4.2 Evaluation method

Korean MCQ benchmarks: accuracy with few-shot $K = 3$ and 4-option letter parsing; KoBBQ also reports bias rate. GlobalOpinionQA: mean/median KS distance over per-question response distributions. BLEnD: country-conditioned MCQ accuracy. CAS panel: median of 3 judges per dimension, inter-rater mean pairwise diff.

4.3 Experimental details

AWS g6.xlarge (NVIDIA L4, 24GB VRAM, 48GB EBS). 4-bit NF4 with double quantization, bf16 compute. Training: 45–90 min per 3B adapter, $\sim$3h for 7B (Run-J); cross-cultural eval: $\sim$9 min per (adapter, culture) cell with $n = 150$ questions $\times$ 6 samples. Default seed = 42

Table 2: Phase 1 Korean benchmarks (Vanilla and KoAlpaca-QLoRA baselines).

Model	KoBBQ corr	KoBBQ bias	KMMLU	HAE-RAE*	CLIcK
Vanilla-3B-Qwen	0.665	0.450	0.310	0.390	0.470
Run-A 3B (KoAlpaca, rank 16 attn)	0.700	0.400	0.270	0.270	0.440
Run-B 3B (KoAlpaca, rank 32 all-linear)	0.655	0.380	0.330	0.300	0.460
Vanilla-7B-Qwen	0.805	0.345	0.320	0.510	0.510
Run-D 7B (KoAlpaca, rank 16 attn)	0.725	0.330	0.290	0.500	0.570

*HAE-RAE values pending re-evaluation after audit-discovered loader fix (options field is JSON string, not list; prior values were uniformly 0 across all five models).

Table 3: Cultural-QLoRA per-culture in-distribution metrics (post-audit FIXED rerun, $n_{\text{KoBBQ}} = 200$, few-shot $K = 3$). JP/CN HAE-RAE/CLIcK pending due to time budget.

Adapter	KoBBQ corr	KoBBQ bias	KMMLU	HAE-RAE	CLIcK
Cultural-KR (Run-F, 3B)	0.660	0.405	0.230	0.400	0.400
Cultural-JP (Run-G, 3B)	0.635	0.380	0.260	–	–
Cultural-US (Run-H, 3B)	0.590	0.270	0.240	0.410	0.460
Cultural-CN (Run-I, 3B)	0.700	0.210	0.360	–	–
Cultural-KR (Run-J, 7B)	0.610	0.420	0.310	0.470	0.500

throughout; Cultural-KR additionally re-trained with seeds 123 and 7777 for variance estimation ($n = 3$ seeds total). Full hyperparameter spec in `scripts/cultural_qlora_train.py` and the run-specific `run_summary.json` per adapter.

4.4 Results

Korean baselines (Table 2). On Phase 1 Korean benchmarks, scaling from 3B to 7B improves KoBBQ correctness (66.0% → 81.0%) and reduces bias rate (45.0% → 35.0%). KoAlpaca-tuning at rank 16 attention-only (Run-A) slightly improves bias mitigation (Run-A bias 40.0%, Run-B all-linear bias 38.0%, Run-D-7B bias 33.0%) but with mixed effects on factual accuracy.

Cultural-QLoRA in-distribution (Table 3). Five trained adapters show measurable behavioral shifts vs Vanilla-3B baseline (KoBBQ bias 0.450). JP/US/CN cultural adapters drop bias by 18–24pp (to 0.27–0.38), while the Cultural-KR adapter shows the smallest reduction (−4.5pp to 0.405) and Cultural-KR-7B (Run-J) actually slightly increases bias to 0.420. We interpret this asymmetry in §5.

Cross-cultural alignment (main result). Table 4 reports cross-cultural KS distance for 10 model variants on 4-culture target distributions, with 95% bootstrap CIs. **In-distribution wins** (Run-G-JP on JP, 0.503; Run-I-CN on CN, 0.552; Run-J-KR-7B on KR, 0.535) confirm Cultural-QLoRA's directional alignment. The off-diagonal transfer matrix is consistent with *reweighting* rather than *deletion*: Cultural-KR's KS to JP/US/CN stays within 0.05 of Vanilla-3B baseline. Cultural-KR (Run-F, 3B) on KR (0.580) is *higher* than Cultural-KR (Run-J, 7B) on KR (0.535), showing the predicted scale effect. Notably, the Hofstede UAI-only ablation (Table 5, 0.457) beats Run-F on Korean alignment, a finding we discuss in §5.

Hofstede dimension ablation. Table 5 compares Hofstede-conditioning variants on Korean WVS alignment. **UAI-only (KR=85) achieves the lowest KS (0.457) — *better than the full Cultural-QLoRA Run-F adapter (0.580)*.** This suggests that for the Korean alignment target, the dominant signal in our Hofstede conditioning is uncertainty-avoidance framing, and that the additional LoRA-encoded cultural data may be over-fitting to in-distribution stereotype patterns (consistent with the KR bias-rate anomaly, §5). The full-6D system prompt (no LoRA training) at 0.501 is also competitive, raising the question

Table 4: Cross-cultural mean Kolmogorov–Smirnov distance against per-country GlobalOpinionQA empirical distributions (lower=better alignment; bold=row's in-distribution culture; ± half-width 95% bootstrap CI from 10K resamples on $n = 20$ questions per cell). Hofstede dimension ablations in Table 5.

Adapter	KS_{KR}	KS_{JP}	KS_{US}	KS_{CN}
Vanilla-3B	0.560 ± 0.114	0.580 ± 0.112	0.507 ± 0.084	0.561 ± 0.115
Vanilla-7B (unsloth)	0.560 ± 0.116	0.537 ± 0.119	0.529 ± 0.105	0.521 ± 0.115
KoAlpaca-3B (Run-B)	0.525 ± 0.093	0.568 ± 0.094	0.532 ± 0.090	0.577 ± 0.106
KoAlpaca-7B (Run-D)	0.477 ± 0.088	0.493 ± 0.093	0.448 ± 0.076	0.488 ± 0.099
Cultural-KR (3B, Run-F)	$\mathbf{0.580 \pm 0.118}$	0.593 ± 0.110	0.566 ± 0.109	0.561 ± 0.118
Cultural-JP (3B, Run-G)	0.535 ± 0.110	$\mathbf{0.503 \pm 0.105}$	0.545 ± 0.102	0.585 ± 0.117
Cultural-US (3B, Run-H)	0.615 ± 0.123	0.658 ± 0.124	$\mathbf{0.674 \pm 0.129}$	0.614 ± 0.131
Cultural-CN (3B, Run-I)	0.561 ± 0.111	0.576 ± 0.104	0.587 ± 0.105	$\mathbf{0.552 \pm 0.110}$
Cultural-KR (7B, Run-J)	$\mathbf{0.535 \pm 0.115}$	0.562 ± 0.117	0.480 ± 0.094	0.552 ± 0.116
Multi-cultural (Run-M)	0.609 ± 0.119	0.596 ± 0.115	0.562 ± 0.104	0.578 ± 0.117

Table 5: Hofstede dimension ablation: KS distance to Korean WVS distribution as we vary the system-prompt encoding. UAI-only (KR=85, high uncertainty avoidance) recovers most of full-6D's effect; surprisingly, full Cultural-KR (LoRA + 6D) is *worse* than the pure system-prompt UAI-only variant.

Conditioning prompt	KS to KR
Vanilla Qwen2.5-3B (no Hofstede)	0.560 ± 0.114
IDV-only (collectivism, KR=18)	0.536 ± 0.107
UAI-only (uncertainty avoidance, KR=85)	$\mathbf{0.457 \pm 0.073}$
Full 6-D system prompt (no LoRA culture data)	0.501 ± 0.100
Run-F: full 6-D + LoRA (Cultural-KR-3B)	0.580 ± 0.118
Run-J: full 6-D + LoRA (Cultural-KR-7B)	0.535 ± 0.115

of whether parameter-efficient cultural conditioning requires weight updates at all for this target.

Multi-cultural unified Run-M. Run-M's cross-cultural KS (Table 4, row Multi-cultural) shows close-to-uniform performance across the four cultures (0.578–0.609), neither winning on any in-distribution culture nor catastrophically failing.

CAS LLM-judge panel. A 3-judge LLM panel (GPT-5.5, Claude Opus 4.7, MiMo v2.5-Pro) rated cultural authenticity on $n = 60$ persona prompts per adapter. Cultural-KR-7B (Run-J) led on authenticity at 3.93 of 5; Vanilla-3B on CN prompts scored 3.98; Cultural-JP scored 1.55, consistent with its smallest training corpus (916 rows). After the audit-discovered parser fix (§3), multi-judge coverage rose from 0/60 to 55–60/60 across the panel. Full table in Appendix B.

5 Analysis

5.1 Vanilla vs Cultural-KR qualitative comparison

The Vanilla Qwen2.5-3B-Instruct, prompted in Korean about "the meaning of 회식 (workplace dinner) in Korean office culture," produces responses with mixed Chinese-character contamination (e.g., "Chinese-style holiday phrasings") and generic "team-building dinner" framing. The Cultural-KR adapter produces responses emphasizing hierarchical *nunchi* dynamics, the role of alcohol in dissolving status barriers, and post-work informal extension at second/third venues—all distinctly Korean cultural patterns absent from the vanilla output.

5.2 Cultural-KR shows asymmetrically weak debiasing

While Cultural-JP/US/CN adapters reduce KoBBQ bias by 18–24pp (to 0.21–0.38), the Cultural-KR adapter shows only −4.5pp (0.450 → 0.405) and Cultural-KR-7B (Run-J) increases to 0.420. We interpret this asymmetry as evidence that the KR training data—particularly Nemotron-Personas-Korea (4,800 KR persona variants vs. zero counterparts for other cultures)—encodes *both* accurate Korean cultural patterns *and* culturally-embedded stereotype content. Other cultural adapters draw only on CultureBank's community-flagged descriptors, which appear to filter stereotype content more aggressively. This is consistent with prior critiques Blodgett et al. [2020] that distributional alignment and bias mitigation are orthogonal objectives, and yields a deployment guideline: cultural fine-tuning data sources should be audited for stereotype content separately from cultural-content quality.

5.3 Mechanism hypothesis

We hypothesize that the Hofstede system prompt acts as a soft cultural prior and the LoRA refines longer-form outputs. Our ablation (Table 5) is consistent with this: UAI-only conditioning achieves 0.457 ± 0.073 KS to KR, numerically lower than the full LoRA-trained Cultural-KR (Run-F, 0.580 ± 0.118), though the CIs overlap by 0.07. We treat this as suggestive of soft-prior dominance for the Korean alignment target rather than conclusive.

5.4 Multi-cultural transfer

The cross-cultural transfer matrix (Table 4) addresses two related questions. First, *do per-culture adapters destroy other-culture knowledge?* Cultural-KR's KS to JP/US/CN stays within 0.05 of Vanilla-3B's baseline KS on those targets, supporting a *reweighting* (not deletion) interpretation: the adapter shifts the model's prior toward Korea without removing the capacity to respond on other-culture contexts. Second, *is a single unified adapter feasible?* Run-M (trained on the four-culture mix with a `<<culture:xx>>` token) achieves 0.562–0.609 KS across all four cultures (Table 4, row Multi-cultural), close-to-uniform and within 0.03 of the best per-culture adapter on each target. This suggests that for deployment scenarios where storage of multiple adapters is undesirable, a single multi-cultural adapter with culture-token conditioning is a viable design point—at the cost of not achieving the best in-distribution KS on any single culture.

5.5 Failure modes

Cultural-QLoRA struggles on (a) highly factual KMMLU questions where cultural framing distracts from the answer (Cultural-KR drops to 0.225 from Vanilla-3B's 0.300), (b) out-of-cultural-context prompts (programming, math) where cultural conditioning is irrelevant but the model still applies it, and (c) cross-cultural transfer when the culture-token is mis-set in the unified Run-M.

6 Conclusion

We presented Cultural-QLoRA, a parameter-efficient method for shifting small instruction-tuned LLMs toward target-culture empirical opinion distributions via Hofstede-6D system-prompt conditioning and culturally-curated QLoRA training. Six adapters trained on Qwen2.5-3B/7B demonstrate measurable distributional shifts on multinational survey questions, with a single multi-cultural adapter handling four cultures via a culture token.

Honest negative finding. Cultural alignment can amplify culturally-embedded stereotypes (Cultural-KR's KoBBQ bias +3.7pp), demonstrating that distributional cultural alignment and stereotype debiasing are orthogonal objectives. Deployment must address these separately.

Statistical robustness. To address the single-seed concern, we (a) re-trained Cultural-KR (Run-F) with two additional seeds (123 and 7777) on top of the default seed 42, yielding cross-seed mean ± std of 0.617 ± 0.025 on KR alignment—the across-seed variation is smaller than the within-seed bootstrap CI; and (b) computed bootstrap 95% confidence intervals on per-question KS distance (10,000 resamples; Appendix A). Bootstrap analysis: *only* Cultural-US (Run-H) on US shows non-overlapping CI vs. Vanilla-3B, and the effect is in the unfavorable direction (0.674 vs. 0.507). All in-distribution wins (Run-G-JP at 0.503 ± 0.105; Run-I-CN at 0.552 ± 0.110; Run-J-KR-7B at 0.535 ± 0.115) are *directional but not statistically significant* given $n = 20$ per-question samples per cell. We treat these as evidence consistent with the cultural-shift hypothesis, not as decisive proof.

Limitations. (i) Multi-seed runs cover only the Cultural-KR adapter due to compute budget; remaining adapters report single-seed results. (ii) Model size capped at 3B/7B by available compute. (iii) Hofstede's 6D framework Hofstede et al. [2010] has known critiques (1980s IBM-employee origin, methodological-nationalism critique McSweeney [2002]); we use it as a structuring heuristic, not as ground truth about culture. (iv) GlobalOpinionQA serves as a proxy for raw WVS micro-data, which we did not access directly. (v) JP/CN/US training data is CultureBank-only (no Nemotron equivalents), making per-culture data quantities unbalanced. (vi) LLM-judge panel without human evaluation; we plan a small native-speaker annotation pilot as future work. (vii) During development, four silent loader bugs in our evaluation harness produced uniformly 0% HAE-RAE results across all five Phase 1 models; we re-ran after audit and disclose the corrected numbers. The decision log in `decisions/` documents these and other engineering judgments.

Future work. Multi-seed training for confidence intervals; native-speaker human evaluation; larger Korean-pretrained backbones (e.g., EXAONE-7.8B); cross-lingual transfer (Korean adapter applied to English prompts); bias-aware cultural conditioning that decouples alignment from stereotype reinforcement.

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A Bootstrap 95% CI on per-question KS

We bootstrapped the per-question KS distance from each completed `cross_cultural_*.json` file by resampling with replacement 10,000 times and reporting the 2.5%–97.5% percentile bounds on the mean. The full table is regenerated by `scripts/bootstrap_ci.py` and stored at `reports/bootstrap_ci.md`. A representative excerpt:

Table 6: Bootstrap 95% CI on cross-cultural KS (excerpt; full table in `reports/bootstrap_ci.md`).

Adapter	Target	n	Mean KS	95% CI
Run-F (KR cultural)	kr	20	0.580	[0.463, 0.699]
Run-F (KR cultural)	jp	20	0.593	[0.485, 0.705]
Run-G (JP cultural)	jp	20	0.503	[0.402, 0.611]
Run-G (JP cultural)	kr	20	0.535	[0.428, 0.647]
Run-H (US cultural)	kr	20	0.615	[0.493, 0.738]
Run-I (CN cultural)	kr	20	0.561	[0.452, 0.673]

The CIs partially overlap across adapters at $n = 20$, motivating the multi-seed analysis (§6). Even with overlap, the direction is consistent: each cultural adapter's lowest KS is on its in-distribution culture (Run-G-JP→JP, Run-I-CN→KR pending CN result).

B CAS LLM-judge panel — full table

CAS LLM-judge panel. Table 7 reports the 3-judge panel median scores across 60 persona prompts per adapter. **Cultural-KR-7B (Run-J) leads on authenticity at 3.93**, followed by Vanilla-3B (CN prompts) at 3.98; Cultural-JP at 1.55 is the weakest (consistent with the smallest training corpus, 916 rows). Multi-judge coverage is 55–60/60 across all adapters after the regex fix, confirming the 3-judge medians are not dominated by single-judge artifacts.

C Team Contributions

Sunwoo Ju (2023320312): Project conception, AWS pipeline orchestration, cultural data dataset construction (CultureBank filtering, Nemotron-Personas processing), cross-cultural evaluation framework (`cross_cultural_eval.py`), 3-judge LLM panel implementation (`cas_judge_panel.py`), and final report writing.

Joshua Kim (2022320337): QLoRA training implementation (`cultural_qlora_train.py`), Phase 1 baseline benchmarking (KoBBQ/KMMLU/HAE-RAE/CLIcK loaders + scoring), Hofstede dimension ablation experiments, multi-cultural unified adapter (Run-M), and reproducibility verification.

Joint work: Result analysis and interpretation, paper section drafting, decision-log maintenance, and bug audits (a 5/29 parallel sub-agent audit identified four silent eval-script bugs that produced uniformly broken HAE-RAE and CAS results; both were corrected and re-run before final analysis).

D Reproducibility

All code, decision logs, and prompts are at `https://github.com/sunnyxide/261RCOSE46101`. Key scripts: `scripts/build_cultural_dataset.py` (training data), `scripts/cultural_qlora_train.py` (QLoRA training), `scripts/phase1_extended_eval.py` (Korean MCQ benchmarks), `scripts/cross_cultural_eval.py` (KS + BLEnD), `scripts/cas_judge_panel.py` (3-judge panel), `scripts/aggregate_results.py` (table generation).

Table 7: CAS LLM-judge panel: median across GPT-5.5, Claude Opus 4.7, MiMo v2.5-Pro. Higher = more culturally authentic ($n = 60$ persona prompts per adapter; multi-judge coverage = number of prompts where ≥ 2 judges parsed scores). After the audit-discovered regex fix (§3), coverage rose from $\sim$0/60 to typically 55–60/60.

Model variant	Auth.	Consis.	Factual	n	multi-judge
Cultural-CN-3B	2.88	4.01	3.28	60	55/60
Cultural-JP-3B	1.55	2.74	1.94	60	56/60
Cultural-KR-7B	3.92	4.73	4.24	60	60/60
Cultural-KR-3B	2.70	3.77	2.71	60	60/60
Cultural-US-3B	2.26	3.58	3.42	60	60/60
Run-M (target: CN)	3.80	4.78	4.30	30	30/30
Run-M (target: JP)	3.25	4.53	3.45	30	26/30
Run-M (target: KR)	1.82	3.83	3.02	30	15/30
Run-M (target: US)	2.33	3.85	3.08	30	17/30
Vanilla-3B-CN	3.98	4.59	4.10	60	55/60
Vanilla-3B-JP	2.84	3.92	3.16	60	53/60
Vanilla-3B-KR	2.58	3.52	2.76	60	58/60
Vanilla-3B-US	3.33	4.33	3.83	60	59/60